

FORECAST-TO-RANK ALLOCATION (F2R)

System Architecture and Operating Framework

Machine-Learning Cross-Asset Strategy with Pretrained Time-Series Intelligence

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Document Summary

This document describes the current Forecast-to-Rank Allocation (F2R) model-portfolio system after the promotion of its heterogeneous forecasting architecture. F2R combines conventional supervised machine learning with Chronos-2 pretrained time-series forecasting, translates the resulting views through a common cross-sectional ranking layer, and maps the final ordering into a disciplined monthly Top-4 model portfolio.

Primary identity: F2R is a governed model-portfolio system. Historical and current-period performance are shown for the current canonical specification applied consistently over the available history. Earlier model versions remain internal research and audit lineage rather than being spliced into the primary performance series. Public materials are architecture-visible and technology-visible while protecting low-level construction parameters that are not required to understand the system.

Terminology

Term	Meaning in F2R v2.1
Current model portfolio	The portfolio generated by the current canonical F2R specification.
Current model history	The current F2R specification reconstructed consistently across valid historical support through the latest available period.
Current portfolio	The current governed monthly model portfolio generated by F2R.
Intramonth Preview	A provisional estimate of the next monthly model portfolio using the latest admissible within-month information.
Chronos-2 pretrained intelligence	A pinned Chronos-2 pretrained time-series forecasting engine used as a distinct sequence expert within the heterogeneous forecasting architecture.
Version lineage	Internal record of earlier model versions, promotion evidence, release receipts, and rollback artifacts.

OVERVIEW

1. Executive Overview

Forecast-to-Rank Allocation (F2R) is a monthly cross-asset portfolio strategy designed around a simple principle: forecasts are useful only after they are converted into a robust portfolio decision. The system therefore separates forecasting from allocation. Multiple forecast engines estimate relative opportunity, the outputs are standardized through a common ranking layer, and the final cross-asset ordering is translated into a Top-4 equal-weight model portfolio.

The current F2R architecture is heterogeneous. Conventional tabular machine-learning models and Chronos-2 contribute structurally different views of the same asset opportunity set. The system does not use raw forecast magnitudes as portfolio weights. Instead, the forecast engines meet at a common rank-score layer, where heterogeneous forecasts are converted into a comparable decision representation before portfolio selection.

What defines the current F2R system

- An 11-ETF cross-asset universe spanning U.S. equities, international equities, real assets, and U.S. Treasury duration.
- A heterogeneous forecasting architecture combining conventional supervised machine learning with Chronos-2 pretrained time-series intelligence.
- Cross-sectional rank consensus before Top-4 selection, rather than direct forecast-proportional portfolio weighting.
- A monthly Top-4 equal-weight model portfolio.
- A point-in-time information clock and simple execution/accounting convention sufficient to preserve model validity.
- Current-model historical performance, MTD, and YTD derived from one internally consistent current-specification path.
- Internal preservation of prior-version evidence without version-splicing the primary performance series.

What F2R is not

- F2R is not an automated brokerage or order-routing system.
- F2R is not a direct forecast-proportional weighting model.
- The public model portfolio is not presented as an audited historical live-account track record.
- Intramonth Preview is not a separate strategy and does not enter the primary historical performance series.
- Earlier model versions do not remain embedded in the current performance history merely because they existed during development.

ARCHITECTURE

2. System Architecture

F2R uses a five-stage architecture: market information, heterogeneous forecast engines, rank consensus, portfolio decision, and governed performance accounting. The architecture is deliberately modular: forecasting technology can evolve while the decision interface remains stable, interpretable, and independent of model-specific output scales.

Stage	Purpose	Current design
1. Market information	Define the point-in-time information set.	Own-price ETF history under the common market clock.
2. Forecast engines	Estimate relative forward opportunity.	Conventional supervised ML plus pinned Chronos-2 pretrained time-series forecasting.
3. Rank consensus	Translate heterogeneous forecasts into a common decision space.	Governed cross-sectional rank consensus with deterministic conflict resolution; exact contribution parameters protected.
4. Portfolio decision	Convert ordering into target positions.	Top-4 equal weight, monthly.
5. Performance accounting	Measure the current model consistently through time.	Daily accounting, drift-aware turnover, and a simple bilateral transaction-cost convention.

Design principle: heterogeneous forecast engines meet at the cross-sectional decision layer. Validation favored rank-based consensus over alternatives that either mixed non-comparable forecast scales directly or combined portfolios only after information had already been discarded. Exact contribution parameters and low-level conflict-resolution rules remain part of the protected implementation layer.

Why heterogeneous engines are useful

The conventional expert summarizes engineered price behavior through supervised learning, while Chronos-2 processes the recent return sequence directly. Their inductive biases differ, so F2R treats them as complementary forecasting experts rather than assuming that one model family should dominate every market regime.

UNIVERSE AND INFORMATION

3. Investment Universe and Information Set

The current universe contains 11 U.S.-listed ETFs spanning major equity, real-asset, and Treasury exposures. The current model uses each asset's own price history; macroeconomic variables are not required.

Asset	Role
SPY	U.S. large-cap equities
QQQ	U.S. growth / technology-heavy equities
IWM	U.S. small-cap equities
EFA	Developed ex-U.S. equities
EEM	Emerging-market equities
EWY	Korean equities
VNQ	U.S. listed real estate
DBC	Broad commodities
GLD	Gold
IEF	Intermediate U.S. Treasuries
TLT	Long-duration U.S. Treasuries

Conventional model information

Information component	Public description
Recent return state	Short-horizon price behavior.
Intermediate trend state	Medium-horizon price trend.
Persistent trend state	Longer-horizon price trend.
Trailing risk state	Recent price variability.
Point-in-time processing	Inputs estimated only inside admissible training data.

Chronos-2 does not consume the conventional tabular feature vector. It receives a governed recent sequence of daily returns, preserving a distinct representation of price history. Exact lookbacks and feature parameters remain protected.

FORECAST ENGINES

4. Forecasting Engines

Conventional supervised expert

The conventional expert combines regularized linear and boosted-tree forecasting models. Each asset is forecast using a time-ordered training process with point-in-time controls. Training observations become eligible only after the return being predicted is fully known, and preprocessing is estimated inside the admissible training sample.

- Ridge regression
- Elastic Net
- Gradient-boosted decision trees (XGBoost)

Pretrained time-series expert

The sequence expert uses a pinned Chronos-2 pretrained time-series model in local offline inference mode. It operates on a governed recent sequence of daily returns and forecasts through the same monthly decision horizon used by F2R. The exact sequence length, runtime parameters, and contribution settings are protected implementation details rather than public portfolio instructions.

Deterministic runtime qualification

- Pinned Chronos-2 revision and validated local artifact identity.
- Offline local model cache; no remote model download is required during normal operation.
- CUDA/GPU runtime qualification on the designated local environment.
- Repeated inference and fresh-reload qualification produced decision-equivalent results under the governed runtime.
- Final Top-4 portfolio identity was stable under observed machine-level numerical differences.

Historical evaluations involving Chronos-2 are classified as retrospective technology-counterfactual evidence. The technology was not historically available at every date in the backtest, so current-model results are not relabeled as a contemporaneous historical live record.

DECISION INTEGRATION

5. Rank Consensus and Portfolio Construction

Forecast engines produce outputs on different numerical scales. F2R therefore converts each engine's forecasts into cross-sectional decision information before combining them. The promoted system forms a governed consensus between the conventional forecasting expert and Chronos-2 at the rank-score layer.

Rank-score integration

- Each engine orders the 11 assets by predicted return.
- Model-specific forecast outputs are converted into cross-sectional rank scores.
- The conventional and pretrained experts are combined under a governed consensus rule. Exact contribution parameters are protected.
- The primary rank construction is deterministic and auditable under the frozen production implementation.
- Residual conflicts are resolved deterministically under frozen implementation rules; the low-level conflict-resolution sequence is not part of the public recipe.
- The final ordering is converted into a Top-4 portfolio with 25% target weight per selected asset.

Why the rank layer is the integration boundary

Architecture testing compared several integration interfaces. The common rank-score layer provided the strongest overall balance of return, risk-adjusted performance, turnover, and stability in the qualified evidence. Public documentation therefore exposes the integration boundary and rationale, while keeping exact contribution parameters and low-level implementation mechanics protected.

Integration interface	Interpretation	Production conclusion
Rank-consensus integration	Heterogeneous experts meet in a common cross-sectional decision space before portfolio selection.	Promoted
Direct forecast-scale blending	Normalized forecast magnitudes are combined directly despite different model scales.	Not selected
Ordinal-only aggregation	Only coarse ordinal positions are combined, discarding relative score information.	Not selected
Portfolio-level blending	Independent final portfolios are combined only after each engine has already selected positions.	Not selected

TIMING AND ACCOUNTING

6. Information Clock, Rebalancing, and Costs

F2R remains a model-portfolio system, not a brokerage simulator. Execution realism is therefore purpose-proportional: the information set, rebalance boundary, turnover, cost, and return timing must be economically valid, while broker routing, partial fills, settlement mechanics, and similar OMS/EMS details are outside the default scope.

Monthly timing contract

- The signal is formed only after the final common U.S. market close of the completed signal month.
- The target is associated with the first subsequent common trading-day close, so the system never assumes execution at the same close that supplied the final signal information.
- The portfolio already in place earns through the rebalance close; the new target begins earning from the next close-to-close interval.
- Between monthly rebalances, weights drift with market returns.

Turnover and transaction-cost convention

Turnover is measured as one half of the L1 distance between drifted pre-rebalance weights and the new target. The current model-history convention applies a simple bilateral transaction-cost rate of 10 basis points per side. This is a model-performance convention designed to prevent unrealistically free rebalancing; it is not intended to reproduce broker-specific fills.

The same timing and accounting convention is used across the current-model historical series, current MTD/YTD, and the downstream PDS constituent history.

CURRENT STATE

7. Current Portfolio, Preview, and Performance Identity

Current model portfolio

The primary portfolio surface shows the monthly portfolio generated by the current canonical F2R specification. The current portfolio is a model decision; it is not presented as a custody or brokerage statement.

Intramonth Preview

Preview applies the current forecasting architecture to the latest admissible within-month market information and shows a provisional estimate of the next monthly model portfolio. It remains separate from the current monthly portfolio and may change before month-end.

Current-model performance identity

F2R follows a current-system performance identity: the latest canonical specification defines the primary model-portfolio history from the beginning of valid support through the latest available period. Cumulative performance, monthly returns, MTD, YTD, annual returns, volatility, Sharpe ratio, drawdown, Calmar ratio, and turnover are derived from that same current-model path.

- No v1-to-v2 performance splice is used in the primary dashboard.
- Earlier F2R versions remain internal research/audit lineage.
- The latest incomplete month is identified as MTD.
- The latest incomplete calendar year is identified as YTD.
- Partial current months are excluded from completed-month hit-rate statistics.

Historical results therefore represent the current governed F2R specification reconstructed consistently over the available support. They are model-portfolio results, not an audited historical live-account track record.

VALIDATION

8. Production Promotion and Validation Evidence

The promoted architecture was selected through staged research and engineering rather than a single headline backtest. Qualification covered portfolio outcomes, execution/cost stress, rolling and regime behavior, numerical determinism, current-state shadow output, and degraded-mode engineering. The public system reports the scope and conclusion of that process without publishing candidate-search recipes or exact integration parameters.

Validation dimension	Result
Headline historical metrics	Promoted candidate improved CAGR, Sharpe, maximum drawdown, Calmar, and turnover relative to the previous current specification over the qualified historical support.
Execution/cost stress	All 12 tested delay-by-cost cells improved the five headline dimensions relative to the prior specification.
Rolling robustness	Every 60-month rolling window in the qualified sample improved all five headline dimensions.
Regime diagnostics	Average active return versus the prior specification was positive in each observed known-at-signal trend/volatility bucket.
Integration robustness	A broad neighborhood of pretrained-expert contribution settings remained strong; promotion was not justified by a single knife-edge parameter choice.
Determinism	Pinned Chronos-2 local runtime repeated inference and fresh-reload tests were decision-stable.
Operational qualification	Dedicated runtime provisioning, current-state shadow, source-owned integration, and local dashboard refresh passed.

Challenger retention

A higher-return/higher-Sharpe challenger architecture remains internal research evidence. It was not selected as the primary production specification because the promoted architecture offered a stronger overall balance of downside behavior, Calmar ratio, turnover, regime breadth, and fallback simplicity. Challenger retention preserves future research optionality without turning the public system page into a candidate-selection recipe.

OPERATIONAL RESILIENCE

9. Runtime States, Failure Handling, and Recovery

F2R separates model/runtime problems from information-integrity problems. This prevents a model failure from being mistaken for valid market information and prevents a data problem from being silently hidden behind a fallback.

State	Meaning	Operational treatment
READY	Current information, conventional models, pretrained runtime, and decision construction pass.	Generate the current model portfolio and Preview.
DEGRADED	The pretrained sequence layer is unavailable but a qualified conventional-only degraded architecture exists.	Use only through explicit controlled recovery/degraded-release authority; do not silently substitute it.
BLOCKED	PIT, source integrity, market data, or another validity gate fails.	Do not issue a new decision.
PREVIEW	Within-month current-model estimate.	Display as provisional only.

Why fallback is not silent

A silent switch to a different model architecture would change the system while preserving the same label. F2R therefore keeps degraded behavior explicit. The normal current-model path fails closed when the promoted pretrained layer cannot be verified, while the qualified conventional-only path remains a separately governed recovery option.

GOVERNANCE

10. Versioning, Reproducibility, and Release Governance

Candidate-to-production promotion

- Research candidates remain separate from production authority until promotion is explicitly approved.
- Promotion freezes the production architecture, runtime identity, information clock, portfolio rule, and public performance identity.
- A candidate freeze preserves evidence lineage but does not permanently freeze the research program.
- Later research may create a new candidate generation without rewriting the evidence supporting the current release.

Reproducibility controls

- Pinned Chronos-2 revision and validated artifact identity.
- Dedicated local runtime separated from research Sandbox state.
- Deterministic rank construction and frozen conflict-resolution behavior.
- Qualified historical target ledger with checksum certification.
- Release manifests and exact-artifact validation before publication.
- Local/current/public state separation and operator review before deployment.

Documentation lifecycle

System Documentation is a living system artifact. A new major documentation version is required when the information set, forecast architecture, rank-integration boundary, universe, portfolio rule, execution/accounting convention, performance identity, or public disclosure boundary changes materially. Editorial refinements that do not alter those authorities are versioned separately.

RELATIONSHIP TO PDS

11. F2R and the Portfolio Decision System

F2R is an independent strategy-level model portfolio. The Portfolio Decision System (PDS) is the portfolio-level integration and monitoring layer. PDS consumes the current F2R constituent history and current F2R portfolio under PDS policy; PDS does not redefine F2R's forecasting architecture.

Dimension	F2R	PDS
Primary role	Generate a forecast-ranked cross-asset model portfolio.	Combine and monitor admitted portfolio-strategy providers.
Current identity	Current canonical F2R specification.	Current canonical PDS composition and orchestration rule.
Historical performance	Current F2R model applied consistently across valid history.	Current constituent histories combined under the current PDS policy.
Current relationship	One of the current PDS Core providers.	Integrates F2R and ADAA in the current Core.
Research ownership	F2R retains its own model-development evidence.	Provider evidence remains provider-owned.

When F2R changes materially, PDS should rebuild its F2R constituent history using the new current F2R specification before recalculating the current PDS Core history. PDS should not preserve an old/new F2R version splice merely because the F2R production version changed.

PUBLIC BOUNDARY

12. Public System Boundary

The public F2R dashboard and SlackQuant system page communicate the current architecture, named technology where it is informative, current model-portfolio state, current-model historical performance, and approved evidence boundary. They intentionally do not expose the full research search process or low-level parameters needed to reproduce the internal construction recipe.

Public	Internal / non-public
Formal F2R identity, named Chronos-2 technology, and high-level heterogeneous forecasting architecture	Exact feature horizons, Chronos-2 context/runtime parameters, integration contribution weights, and low-level conflict-resolution rules
Current model portfolio and Intramonth Preview	Local paths, runtime caches, environment details, and private state
Current-model historical performance and target history	Superseded model-version chronology and rollback state
High-level validation status and evidence boundary	Full hostile-review record, candidate-search artifacts, and non-promoted challenger specifications
System Documentation and related research links	Unpromoted experimental branches

Public architecture language

Market Information → Conventional ML + Chronos-2 → Rank Consensus → Portfolio Decision → Governed Performance

External resource map

Resource	Location
F2R System page	research.slackquant.com/systems/f2r/
F2R Public Dashboard	f2r-forecast-to-rank-allocation.streamlit.app
System Documentation	research.slackquant.com/resources/systems/f2r/F2R_System_Documentation_v2.1.pdf
Related Research	research.slackquant.com/research/price-macro-decision/
Public deployment repository	github.com/slackquant80/f2r-forecast-to-rank-allocation
Quantitative Methods	research.slackquant.com/methods/

REFERENCE

13. Current Specification Summary

Item	F2R v2.1
Public name	Forecast-to-Rank Allocation (F2R)
System type	Portfolio Strategy System
Universe	11 U.S.-listed ETFs
Information set	Own-price ETF history
Conventional forecast expert	Regularized linear models + gradient-boosted trees
Pretrained forecast expert	Pinned Chronos-2 pretrained time-series model
Integration	Governed cross-sectional rank consensus; exact contribution parameters protected
Portfolio rule	Top-4 equal weight
Decision cycle	Monthly
Timing	Signal after completed month-end common close; next common close is the rebalance boundary
Weight evolution	Market-driven drift between monthly rebalances
Turnover	Half-L1 versus drifted pre-rebalance weights
Cost convention	10 bp per side, bilateral model convention
Primary performance	Current canonical model reconstructed consistently through the latest available period
MTD / YTD	Same current-model performance path
Preview	Intramonth provisional current-model estimate
Historical version handling	Prior versions retained internally; no primary current-history splice
PDS relationship	Independent strategy provider consumed by the current PDS Core

About this version

Version 2.1 is an editorial and disclosure-boundary refinement of the promoted current F2R architecture. It makes Chronos-2 and the heterogeneous forecasting design explicit while protecting exact feature parameterization, sequence settings, contribution parameters, and low-level conflict-resolution mechanics. No model logic, portfolio decision, performance history, execution/accounting convention, or PDS relationship changes from v2.0. Version 2.1 supersedes v2.0 as the current public System Documentation; earlier versions remain historical records.